

Lecture 2

Unit 1 Time Series

By Sanchit, Rishabh and Shobhit



Join the lecture online on *your* dashboard.

Let's start with a minute of silence.

आचार्यात् पादं आधत्ते पादं शिष्यः स्वमेधया । पादं सब्रह्मचारिभ्यः पादं कालक्रमेण च ॥

Meaning:

A student acquires knowledge in four equal parts:

- One-fourth from the teacher
- One-fourth through self-reflection and independent thinking
- One-fourth through discussions with peers and fellow learners
- One-fourth over time through personal experience



Recap

- **Time Series**

- Time series: Data points recorded over time (in time order).
- What is NOT a time series: Data without a time order (e.g., customer ages, marks list, product prices without dates).

- **Core Concepts / Components**

- Time Index: The timestamp/date showing when each value occurs.
- Frequency: How often values are recorded (e.g., hourly, daily, weekly).
- Trend: The long-term direction of the series (upward/downward/stable).
- Seasonality: Repeating patterns at fixed intervals (daily/weekly/yearly cycles).
- Noise: Random variation that cannot be explained by pattern/trend.

- **Decomposition:** Breaking the time series into parts (Trend + Seasonality + Noise) to visualize and understand it better.

Agenda

- Stationarity vs. Non-Stationarity
- Differencing Transformations
- ACF (Autocorrelation Function)
- PACF (Partial Autocorrelation Function)

A Curious Question Before We Begin...

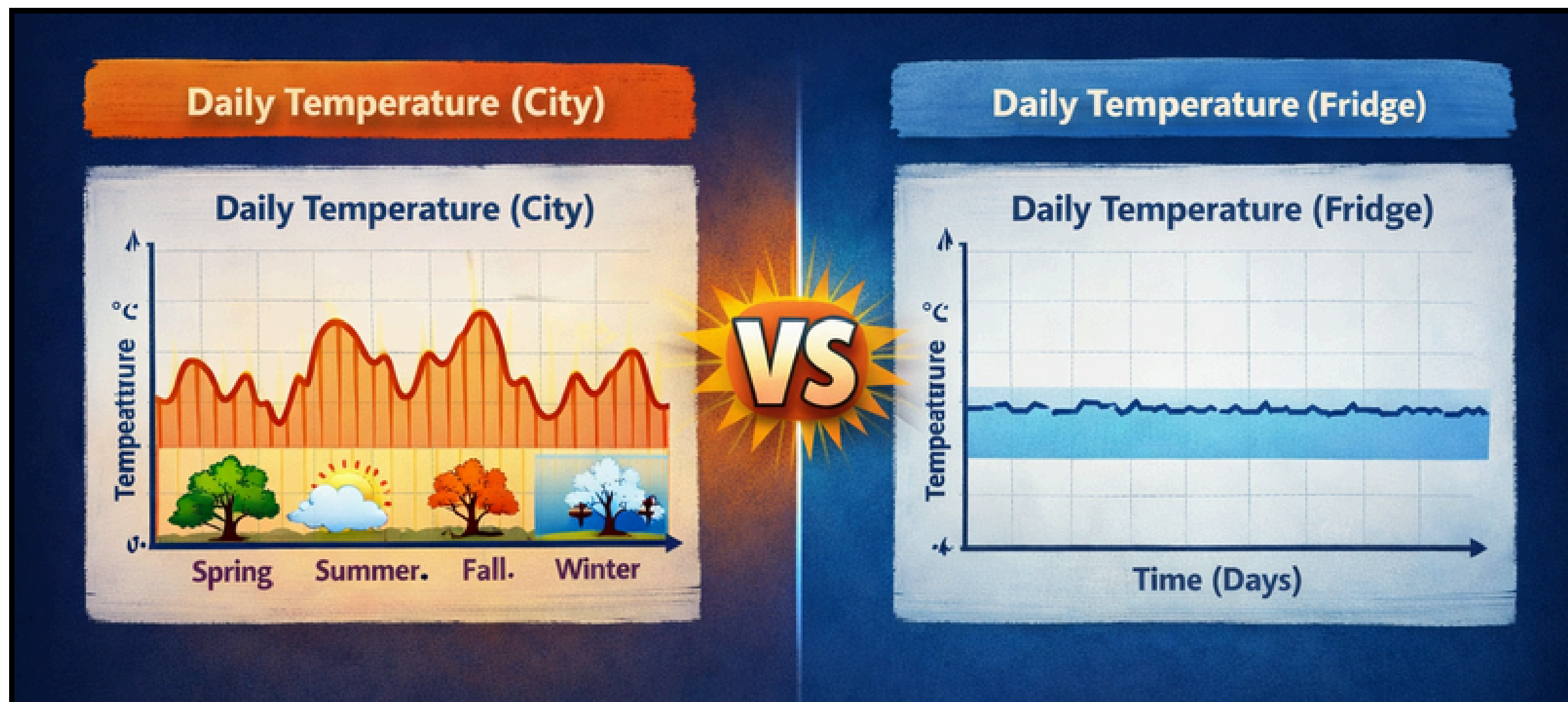
- **Case 1:** Imagine you measure the **daily temperature** in your **city** for a **year**.
- **Case 2:** Now imagine you also measure the **daily temperature inside** your **fridge** for a **year**.

**Which one would be easier to predict using past data,
and why?**

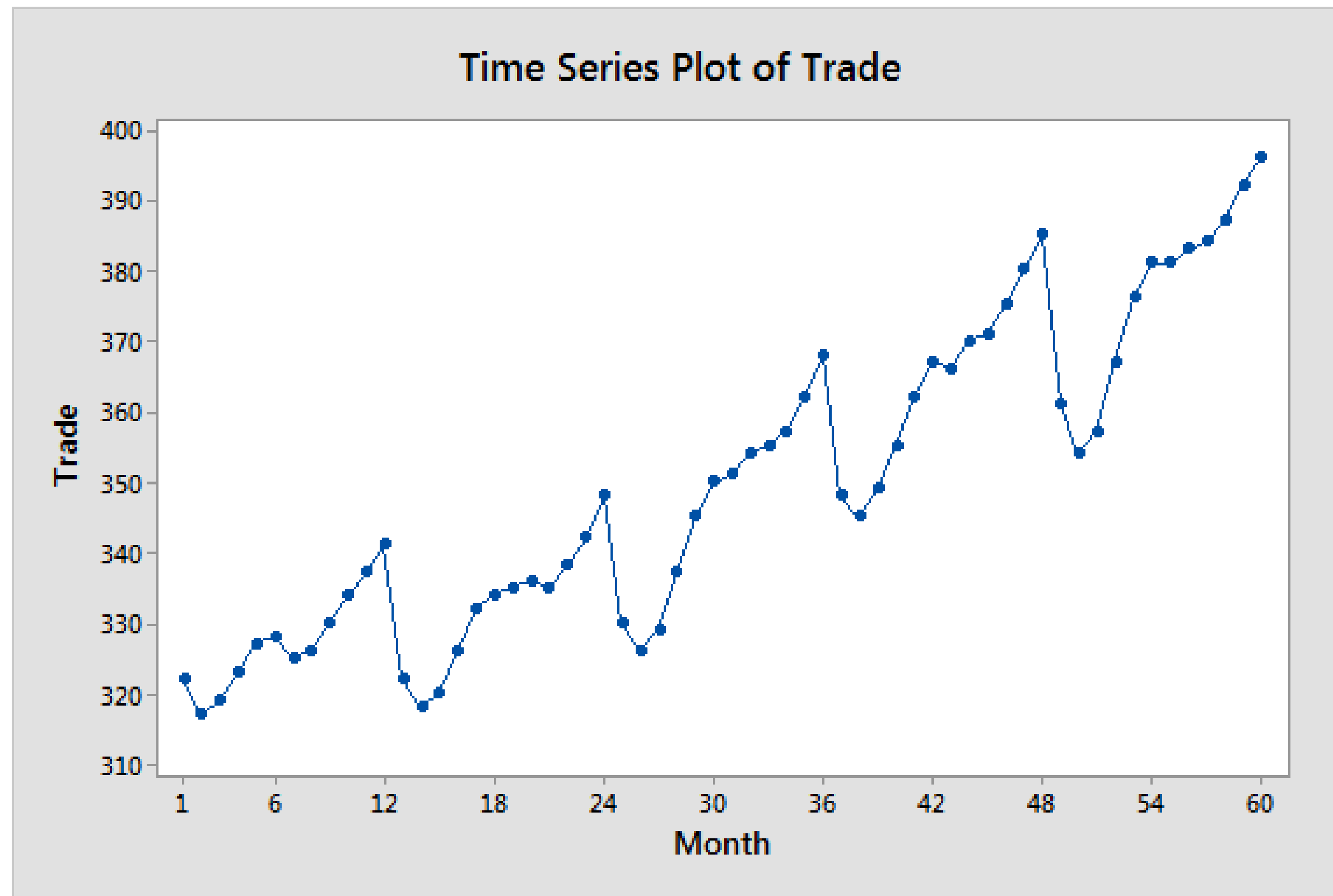
A Curious Question Before We Begin...

- **Case 1:** Imagine you measure the **daily temperature** in your **city** for a **year**.
- **Case 2:** Now imagine you also measure the **daily temperature inside** your **fridge** for a **year**.

Which one would be easier to predict using past data, and why?

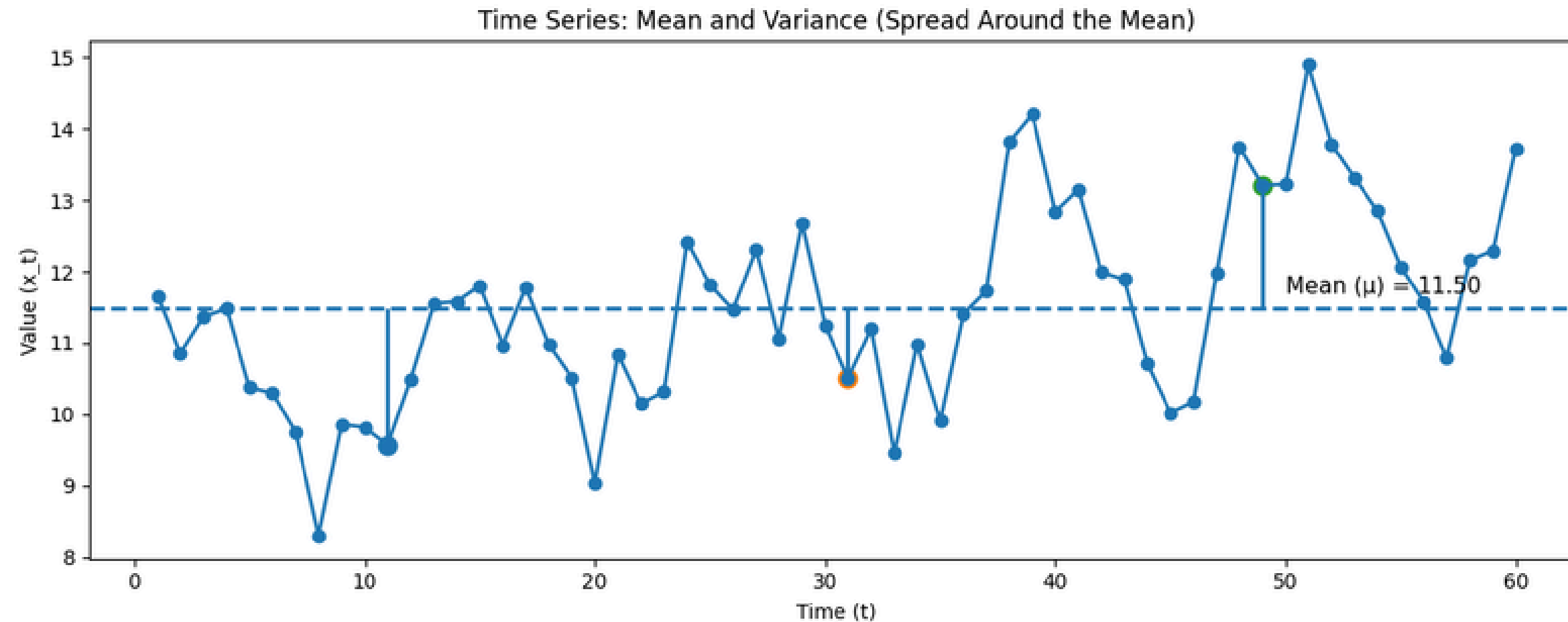


- **Fridge temperature:** fluctuates, but around a constant level.
- **City temperature:** changes across different months, day and night.



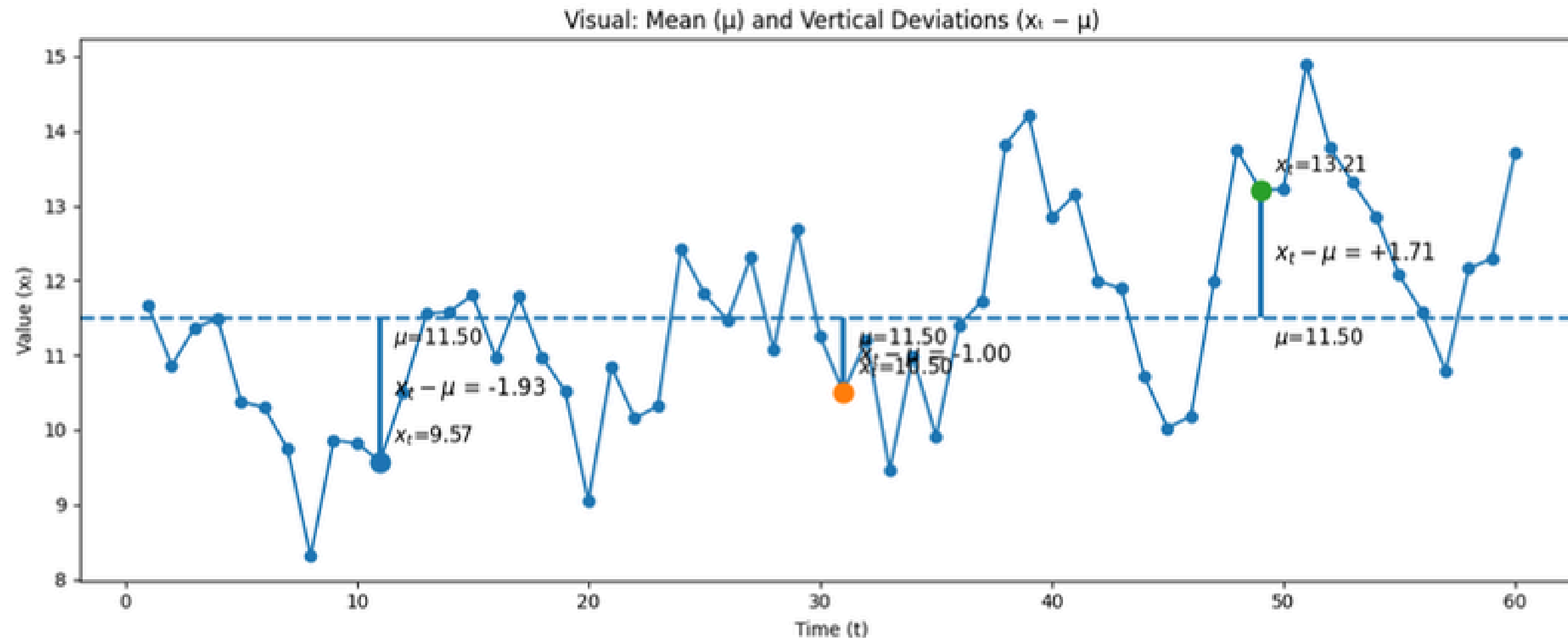
What is mean, variance and correlation in timeseries?

Mean and Variance in Time Series



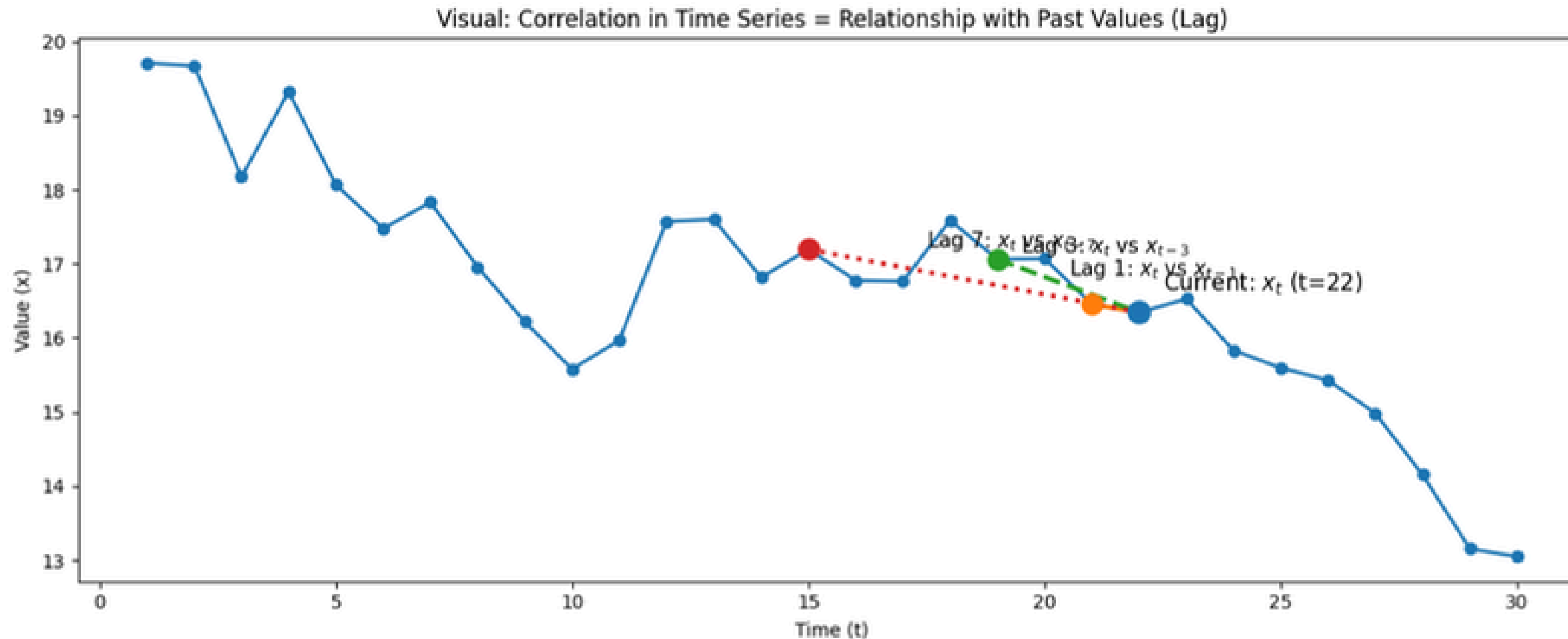
- Dashed horizontal line = Mean (μ)
- Vertical lines from mean to points = deviation from mean (these deviations create variance)

Mean and Variance in Time Series



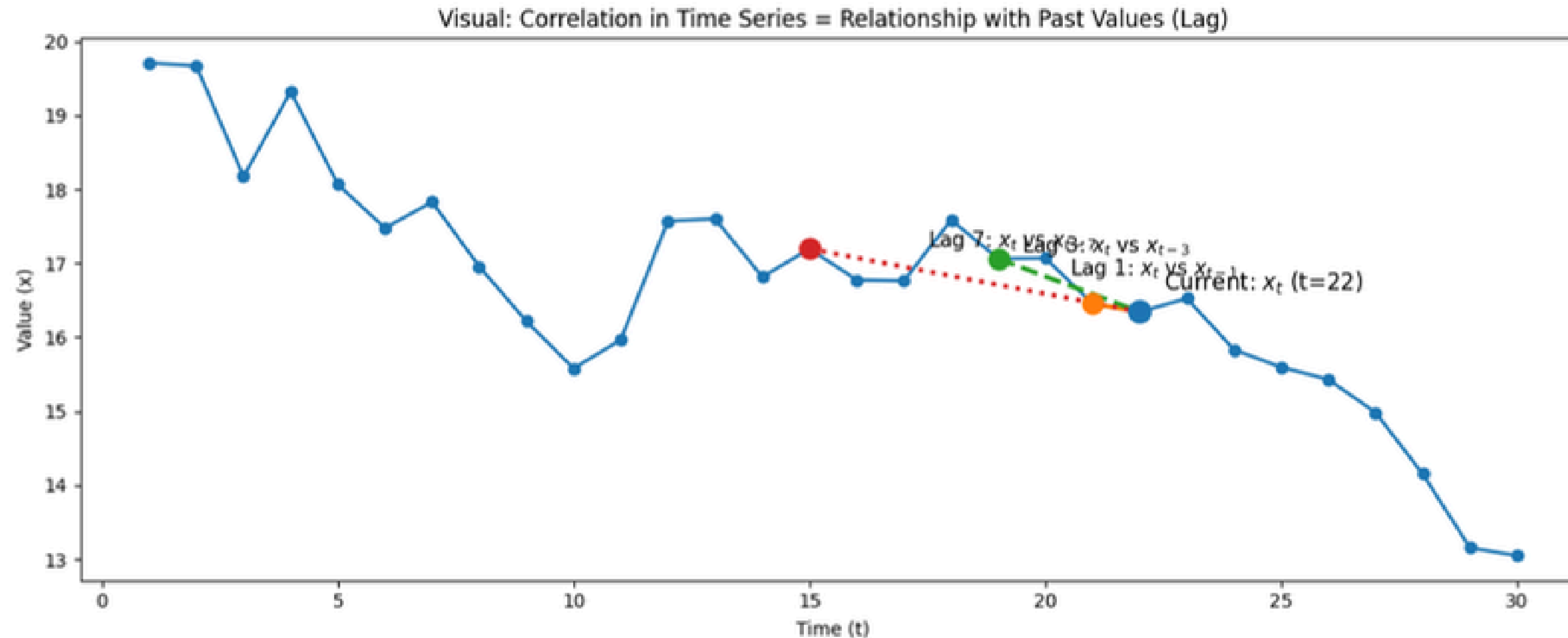
- Mean line (μ) = “average level” of the series
- A point $\mathbf{x(t)}$ = actual value at time t
- Vertical line length = $\mathbf{x(t)} - \mu$
- This is called **deviation from mean**.

Correlation in Time Series



- What is correlation? How strongly **two values** move together.
- In time series, we don't compare two different **variables**, we compare the same **variable** at **two different times**.

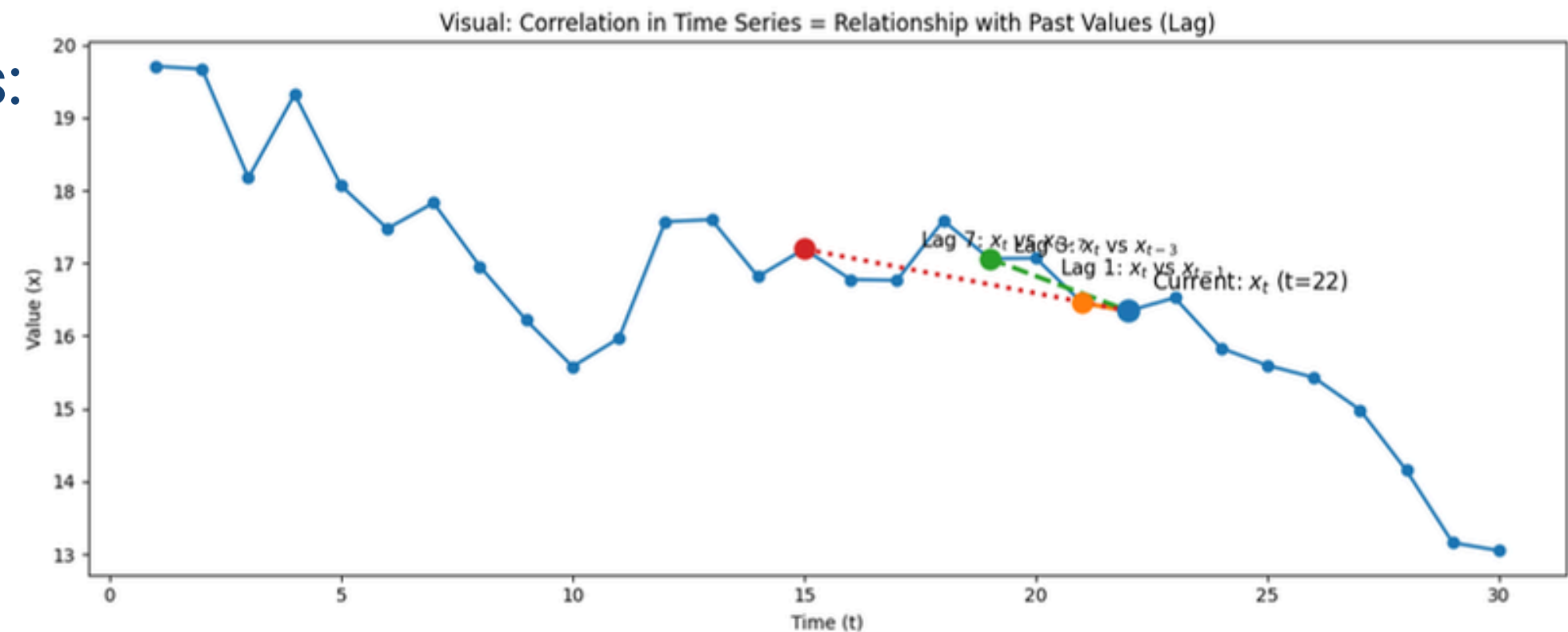
Correlation in Time Series



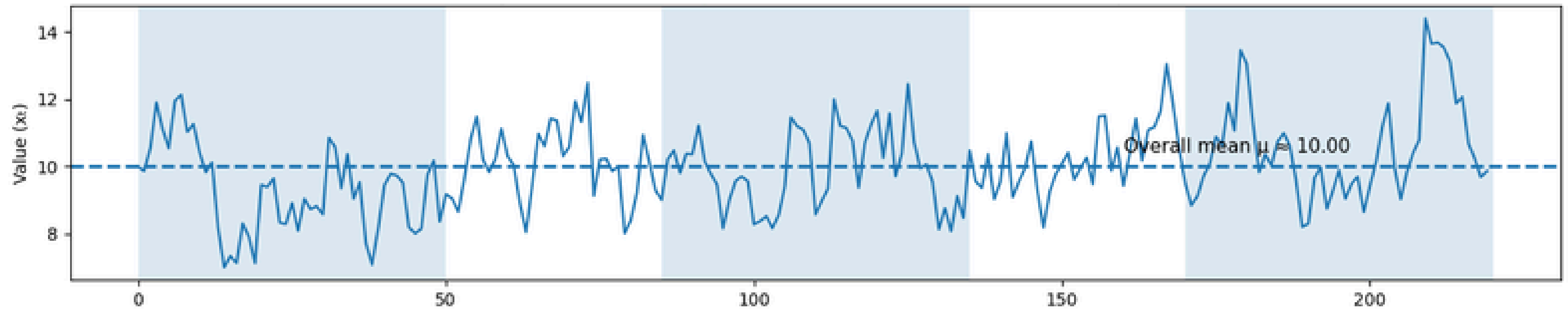
- How to calculate correlation in timeseries?
- **We will cover this in depth in upcoming slides.**

Correlation in Time Series

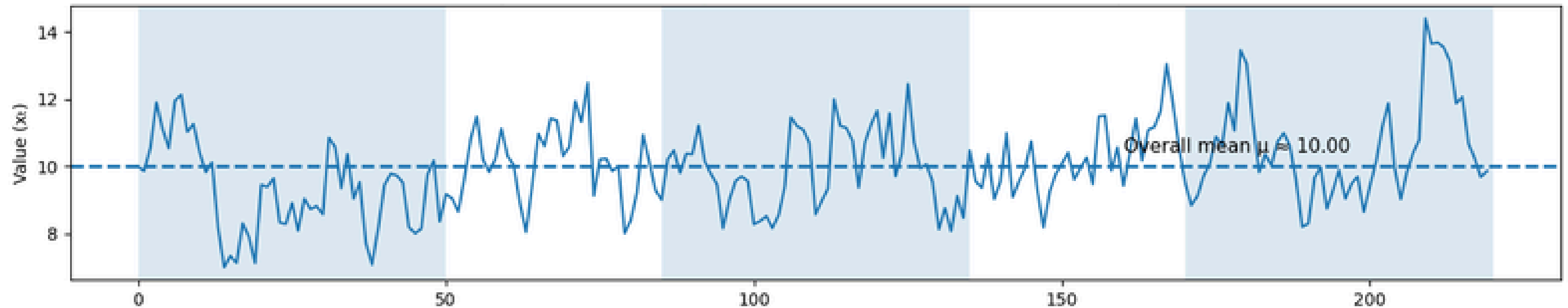
- How to calculate correlation in timeseries? with **Lags**
- The **blue dot** is the **current value $x_{(t)}$**
- The connected dots show past values:
 - **Lag 1:** compare $x_{(t)}$ with $x_{(t-1)}$
 - **Lag 3:** compare $x_{(t)}$ with $x_{(t-3)}$
 - **Lag 7:** compare $x_{(t)}$ with $x_{(t-7)}$
 - **Lag K:** compare $x_{(t)}$ with $x_{(t-k)}$



If points that are connected (lagged points) are usually close / move similarly, the correlation is high → past helps prediction.



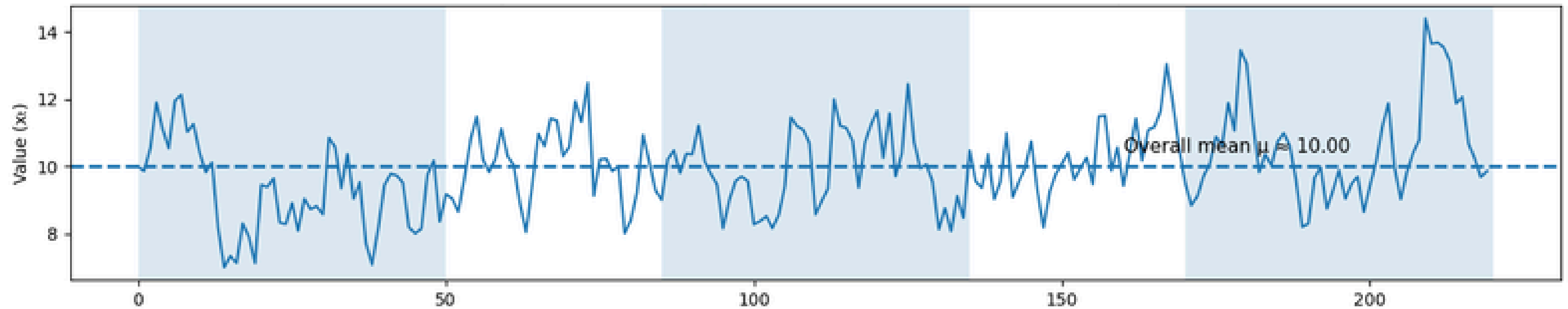
What is your **observation** about this **timeseries**?



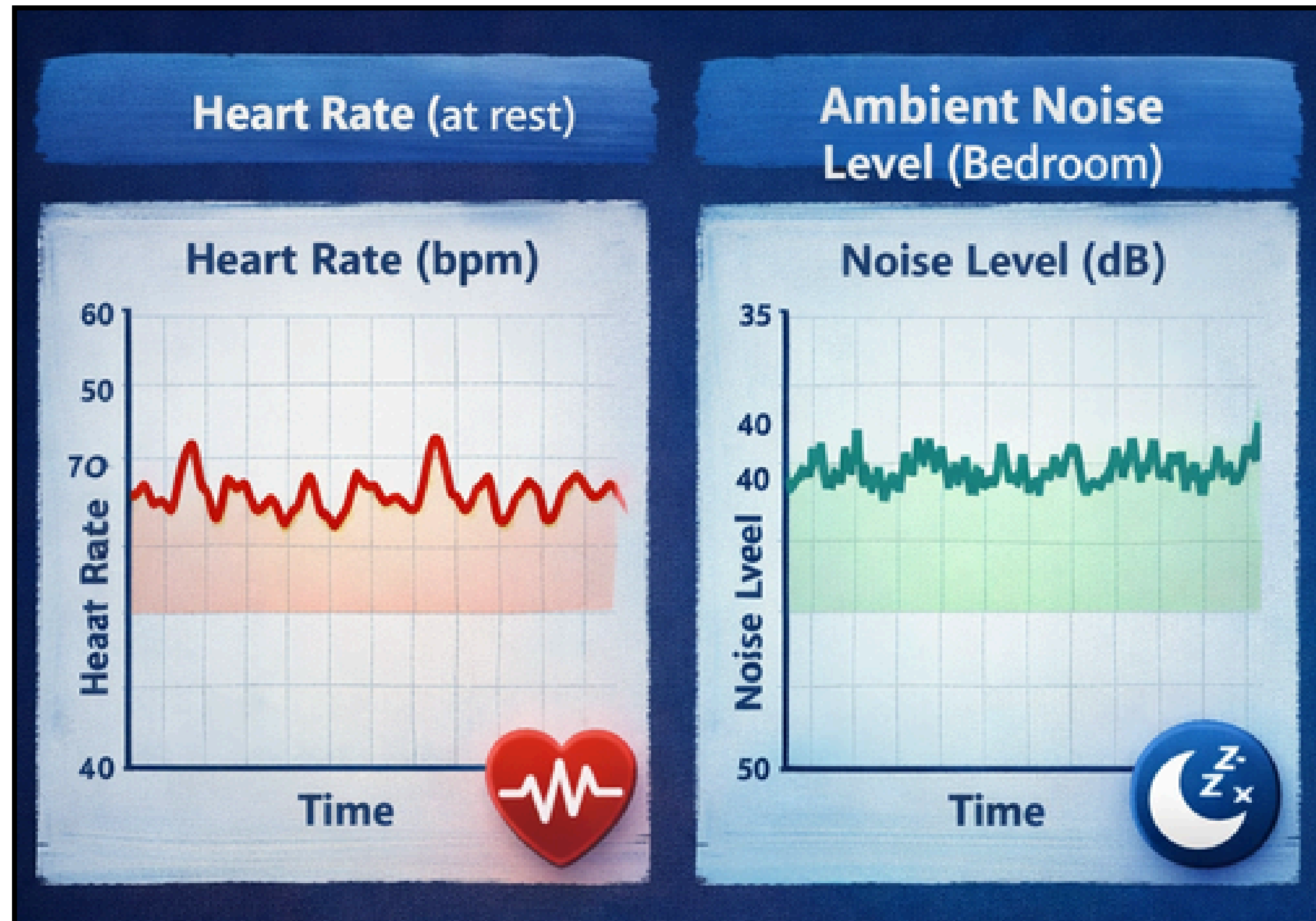
What is your **observation** about this **timeseries** **specifically around mean and variance??**

- A time series is stationary if its **behavior** does not **change over time**.
- Which means if its **major statistical properties (Mean, variance, etc.)** remain constant over time.
- A time series is stationary when:
 - **Mean** is constant (average level stays same)
 - **Variance** is constant (spread/volatility stays same)
 - **Correlation** depends only on **lag**

Is this time series Stationary?

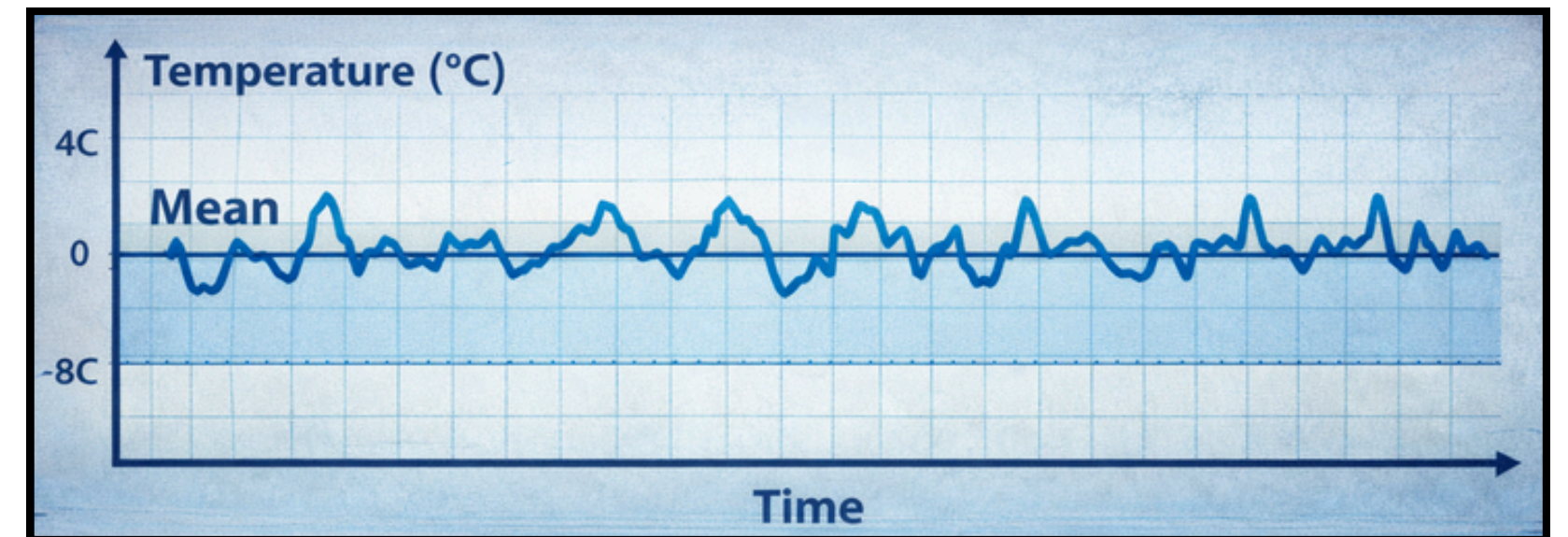


Real life examples of Stationary Time Series Data



Mean Stability

A stationary series should “hover” around the **same average** level at all times this is called as **Mean Stability**.



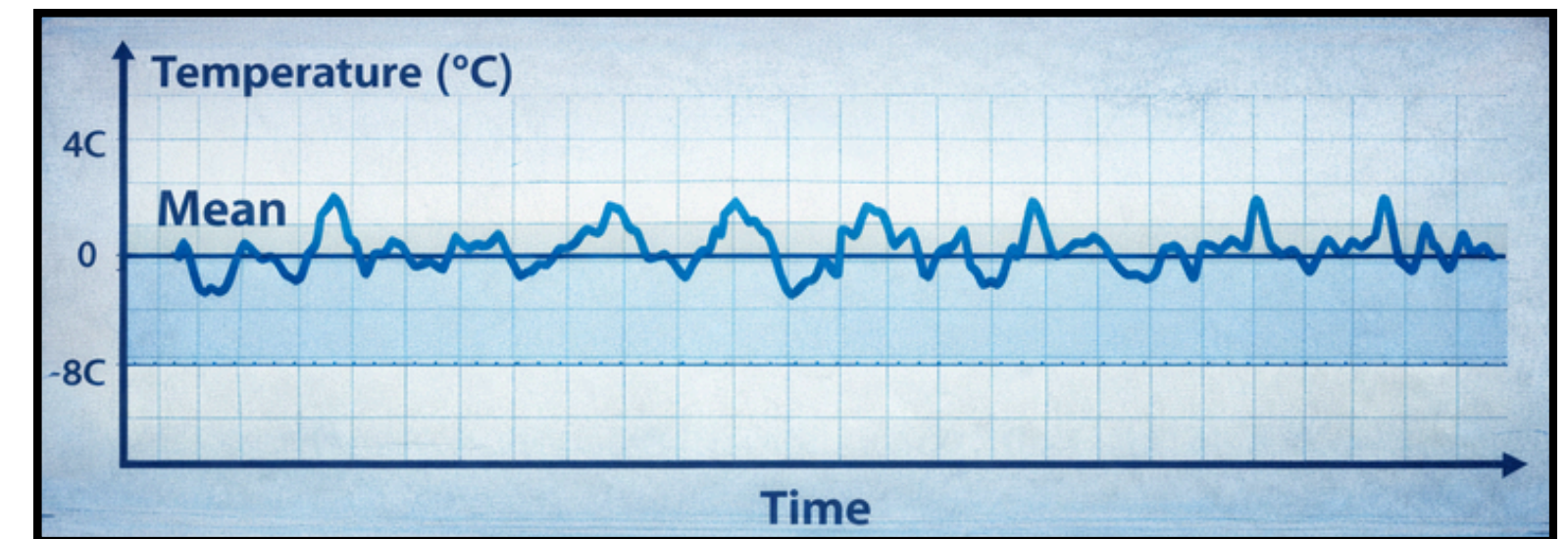
Mean Stability

Think about a fridge:

- It is set to a fixed cold temperature.
- Even if it fluctuates a little when you open the door,
- It always returns to roughly the same level.

The average temperature stays the same every day.

This means the **fridge temperature** has a **mean stability**.

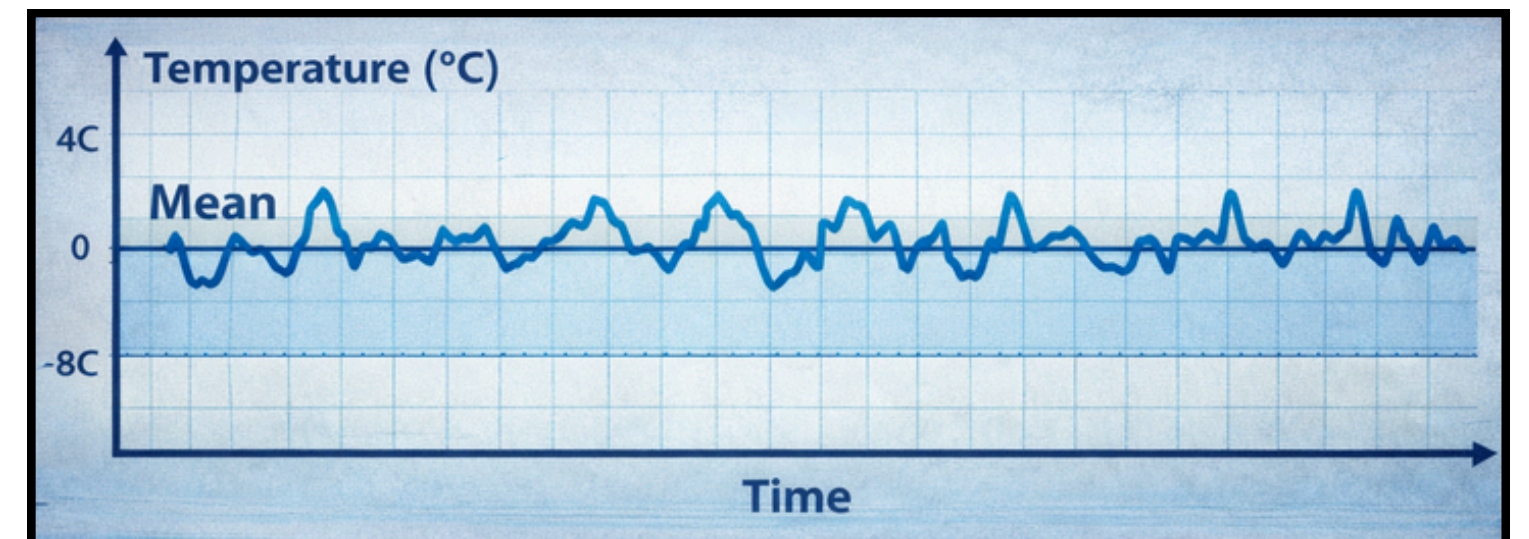


Variance Stability (Constant Variance)

A stationary series keeps its spread or variance **consistent**.

The variance stays the same every day.

This means the fridge temperature also has **variance stability**.



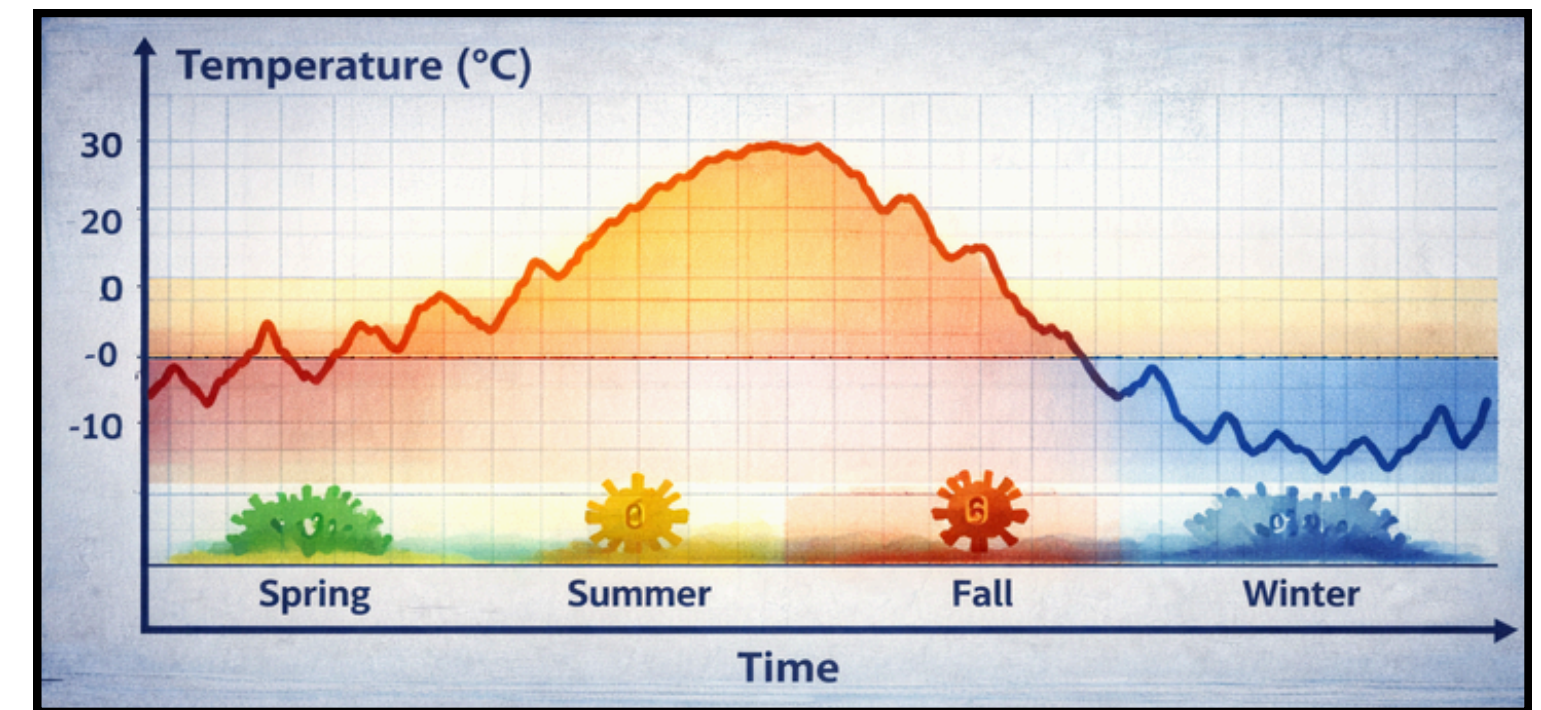
Examples of Non-Mean Stability

Now look at outdoor temperature:

- In summer, the average temperature is higher.
- In winter, the average temperature is lower.
- The typical temperature changes through seasons.

The average does **not stay the same**.

The **outdoor temperature** does not **have mean stability**.



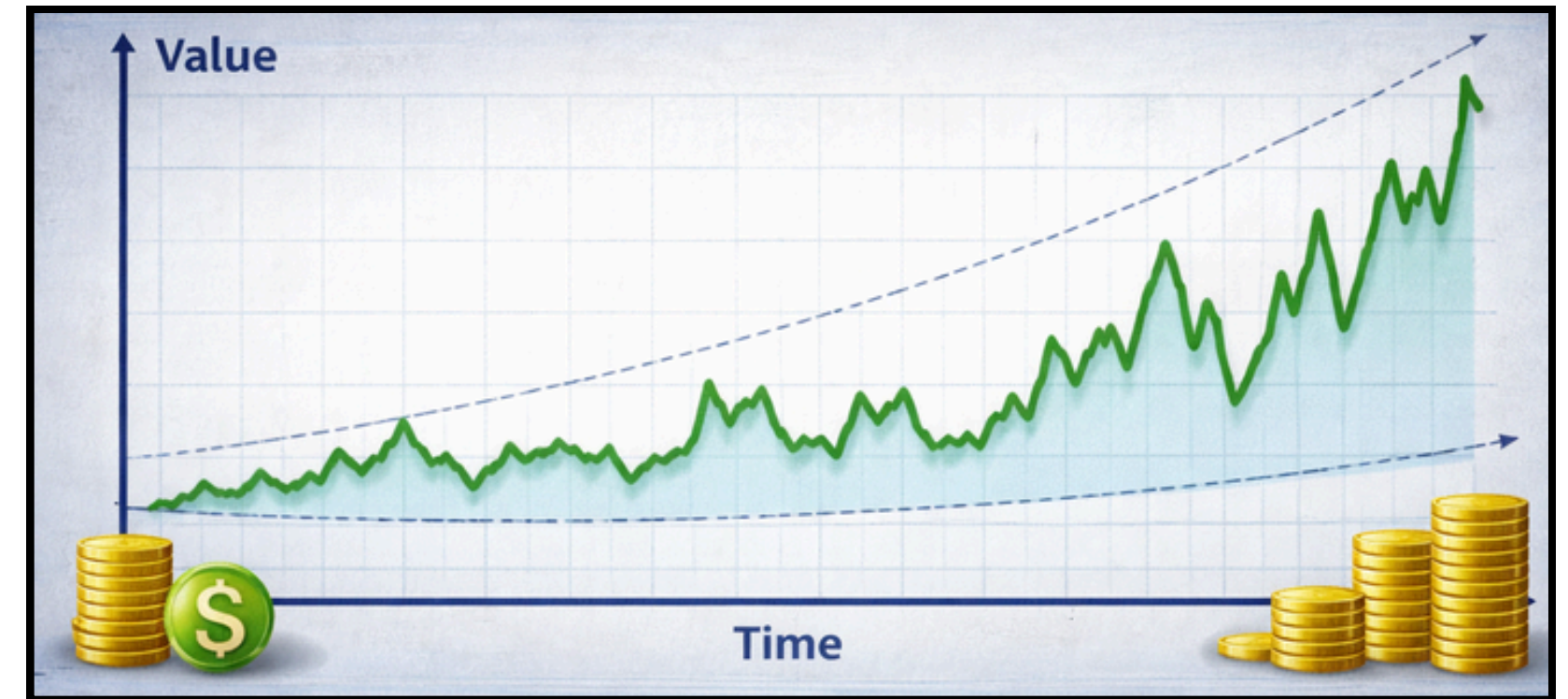
Example of Non-Variance Stability

Non-variance stability patterns include:

- increasing volatility
- sudden jumps
- variance that expands over time

Example:

Financial markets often show this: small fluctuations early, large swings later.



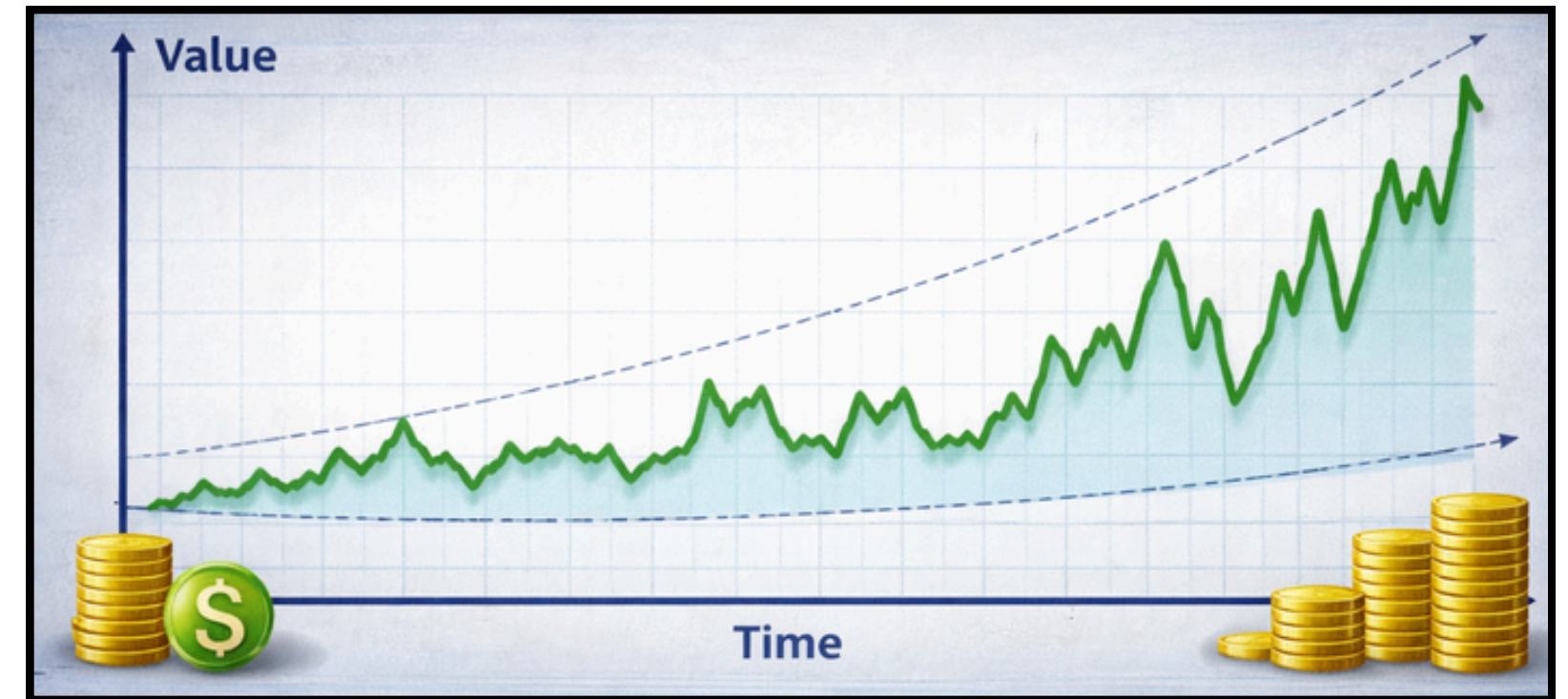
Non Stationary Data

Non-stationary data patterns include:

- upward trend
- downward trend
- structural breaks
- seasonality

Example:

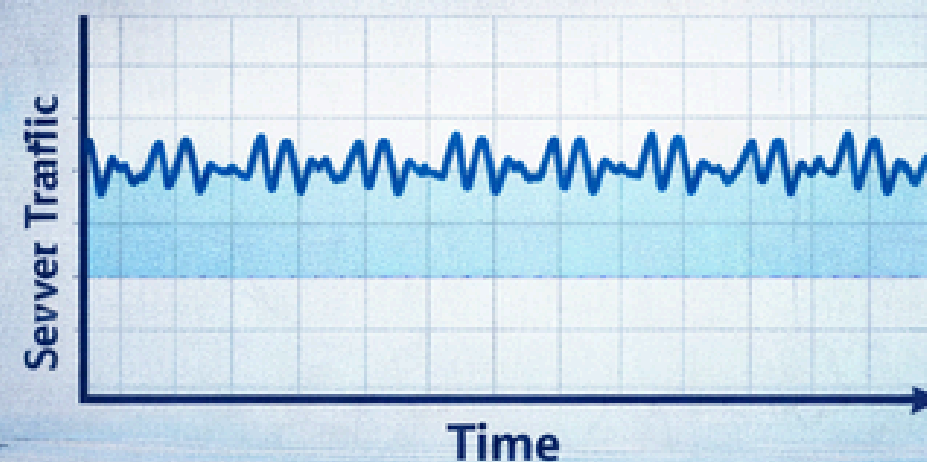
Monthly sales are steadily rising → the mean is changing.



Identify the following graphs as Stationary or Non-Stationary

Real-Life Example Graphs: Stationary or Non-Stationary?

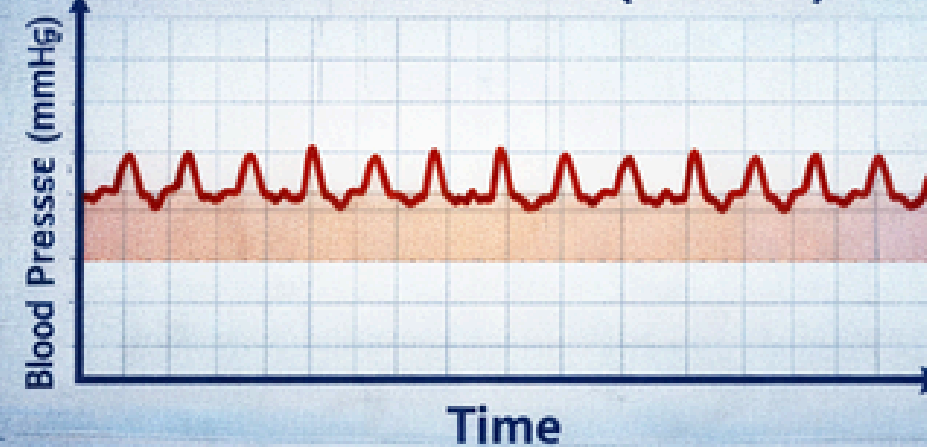
Online Game Server Traffic



Daily COVID-19 Cases



Blood Pressure (at rest)



Size of Ice Cap (Climate)



Why Stationarity Matters?

Few Reasons why Stationarity matters in time series data:

- Past helps predict the future
- Consistent behavior
- Reliable predictions
- Simpler analysis

Stationary vs Non-Stationary Data

Aspect	Stationary	Non-Stationary
Mean	Constant	Time-dependent
Variance	Constant	Changes
Modeling	Easier	Harder
Forecasting	Stable	Risky
Real-world realism	Lower	Higher
Statistical tests	Valid	Often invalid

Problem Statement

If a restaurant received **X Zomato orders today**, can we predict **tomorrow's orders** using just that **single number**?

Or do we need to know what happened **yesterday**, the **day before**, or **last week**?

- **Food orders usually don't behave randomly. There are predictable effects:**
 - If **Monday** is busy → **Tuesday** is generally similar.
 - **Weekend** orders are consistently higher.
 - **Rainy** days often **increase orders** for 2–3 days.
 - A **discount** running today affects **orders** tomorrow.
- **Today's order volume is related to yesterday's order volume.**

Day	Orders	Contextual Factor
Monday	240	Regular weekday
Tuesday	255	Exam season begins
Wednesday	265	Exam season continues
Thursday	260	Light rain
Friday	275	Rain + pre-weekend

Lag Dependence (How Far Back the Relationship Goes)

- **Lag means:** how many time-steps back we look.
 - So if today is Day T, then lag tells us which previous day/value we are using.
- If we are looking at today's value:
 - **Lag 1 → yesterday**
 - **Lag 2 → day before yesterday**
 - **Lag 7 → same day last week**
 - **Lag 30 → same day last month**
- **Basically: Lag = past value used to understand today.**

Day	Orders	Orders (Lag 1)
Monday	240	
Tuesday	255	240
Wednesday	265	255
Thursday	260	265
Friday	275	260

Lag Dependence (How Far Back the Relationship Goes)

Let the time series be: y_t

Where:

- t = Current time
- y_t = value at time t

Then, **Lag-k** value is: y_{t-k}
meaning the value at time k units before
time t i.e. value at time $t-k$.

- **Think of a lag as a conversation between two time points.**
 - “What happened today?”
 - “What happened one step before?”
- If today and yesterday are similar → strong lag-1 relationship.
- If today and **7 days** ago are **similar** → **strong lag-7 relationship**.

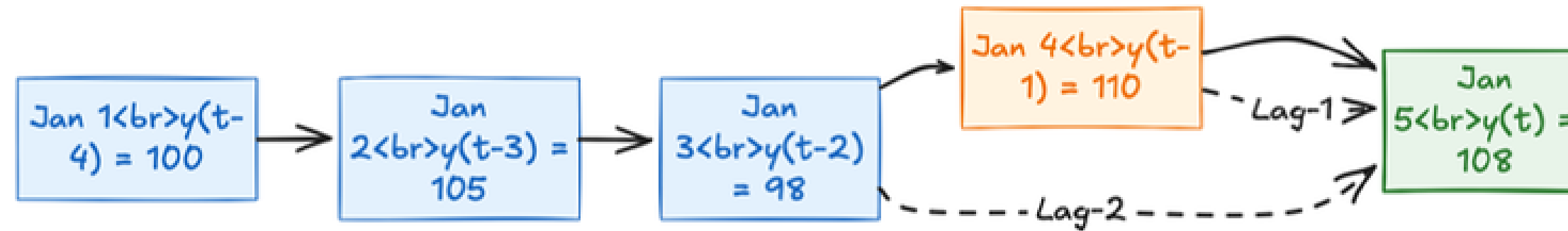
Lag Dependence

Day	Orders	Lag 1	Lag 2	Lag 7	Comments
Mon (Week 1)	220	—	—	—	Start
Tue (Week 1)	230	220	—	—	Slight weekday rise
Wed (Week 1)	235	230	220	—	
Thu (Week 1)	240	235	230	—	
Fri (Week 1)	300	240	235	—	Weekend spike
Sat (Week 1)	320	300	240	—	
Sun (Week 1)	310	320	300	—	
Mon (Week 2)	225	310	320	220	Weekly pattern
Tue (Week 2)	235	225	310	230	

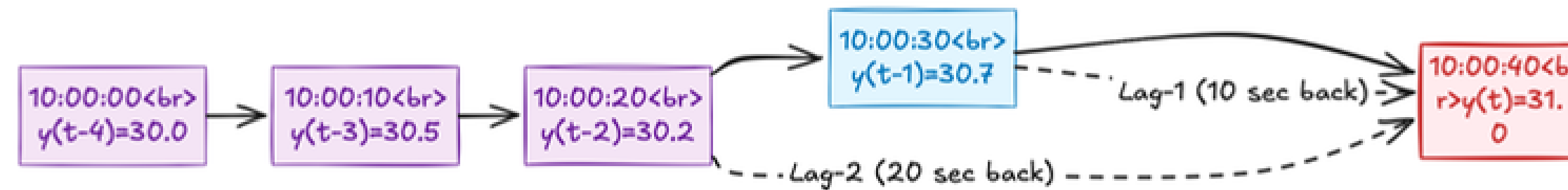
Lag = “how many steps back” (step size depends on granularity)

Data Type	1 step =	Lag 1 means
Second-wise	1 second	1 second ago
Minute-wise	1 minute	1 minute ago
Hourly data	1 hour	1 hour ago
Daily data	1 day	1 day ago

Lag for Daily Data (Lag-1, Lag-2)



Lag for Time Data (every 10 seconds)



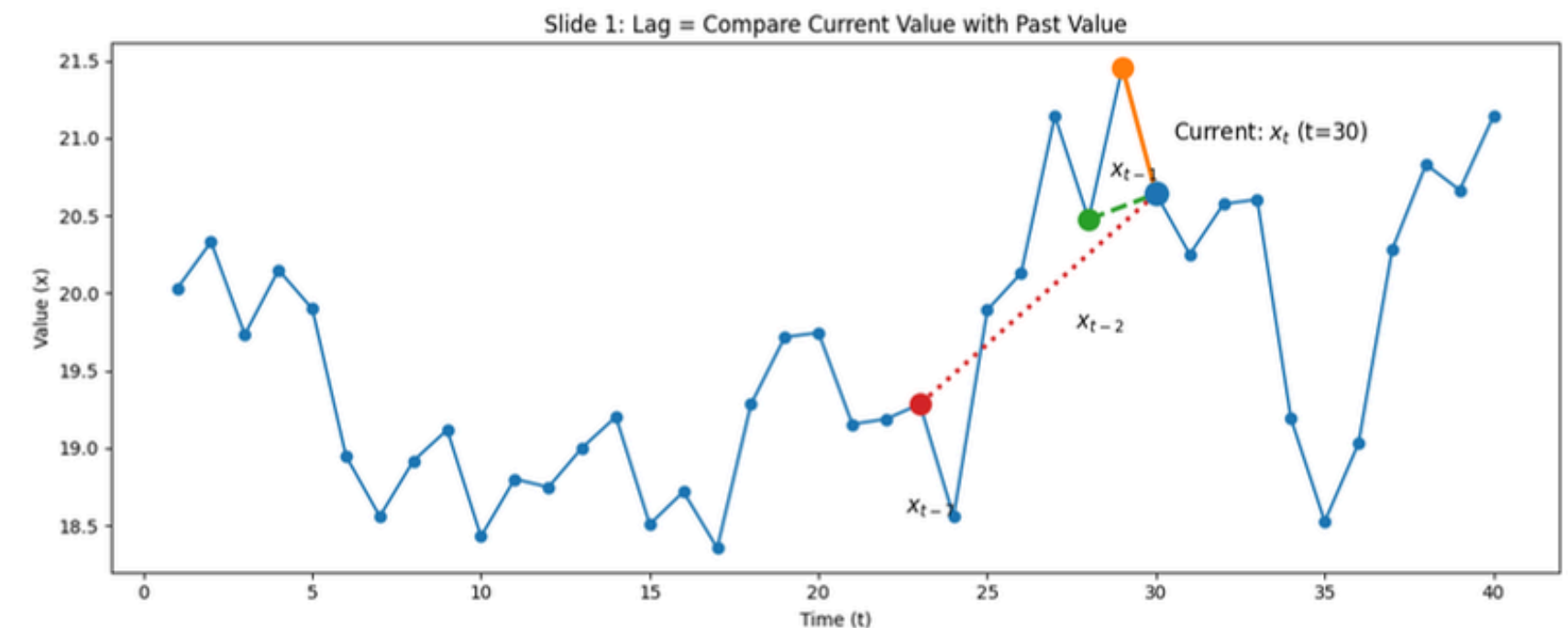
How do we choose useful lags?

Use lags based on **domain pattern**:

- **Daily series:**
 - Lag-1 (yesterday effect)
 - Lag-7 (weekly seasonality)
- **Hourly series:**
 - Lag-1 (last hour)
 - Lag-24 (same hour yesterday)
- **15-min series:**
 - Lag-1 (last 15-min)
 - Lag-96 (same time yesterday)

Practice rule:

Pick lags where **real-world cycles exist**.



Problem Statement

Is this data stationary?

Time (t)	Value
1	5
2	7
3	9
4	11
5	13
6	15
7	17

Problem Statement

This data is **not stationary**, as there is a clear upward trend in the data with respect to time.

Time (t)	Value
1	5
2	7
3	9
4	11
5	13
6	15
7	17

Problem Statement

How would you
convert this data
into stationary
data?

Time (t)	Value
1	5
2	7
3	9
4	11
5	13
6	15
7	17

Problem Statement

We will apply the
transformation in the
data

Time (t)	Value
1	5
2	7
3	9
4	11
5	13
6	15
7	17

Difference Transformation

We take the difference of the two lags to make the data stationary.

$$\Delta X_t = X_t - X_{t-1}$$

X_t : **Current Value**

X_{t-1} : **Previous Value (Lag 1)**

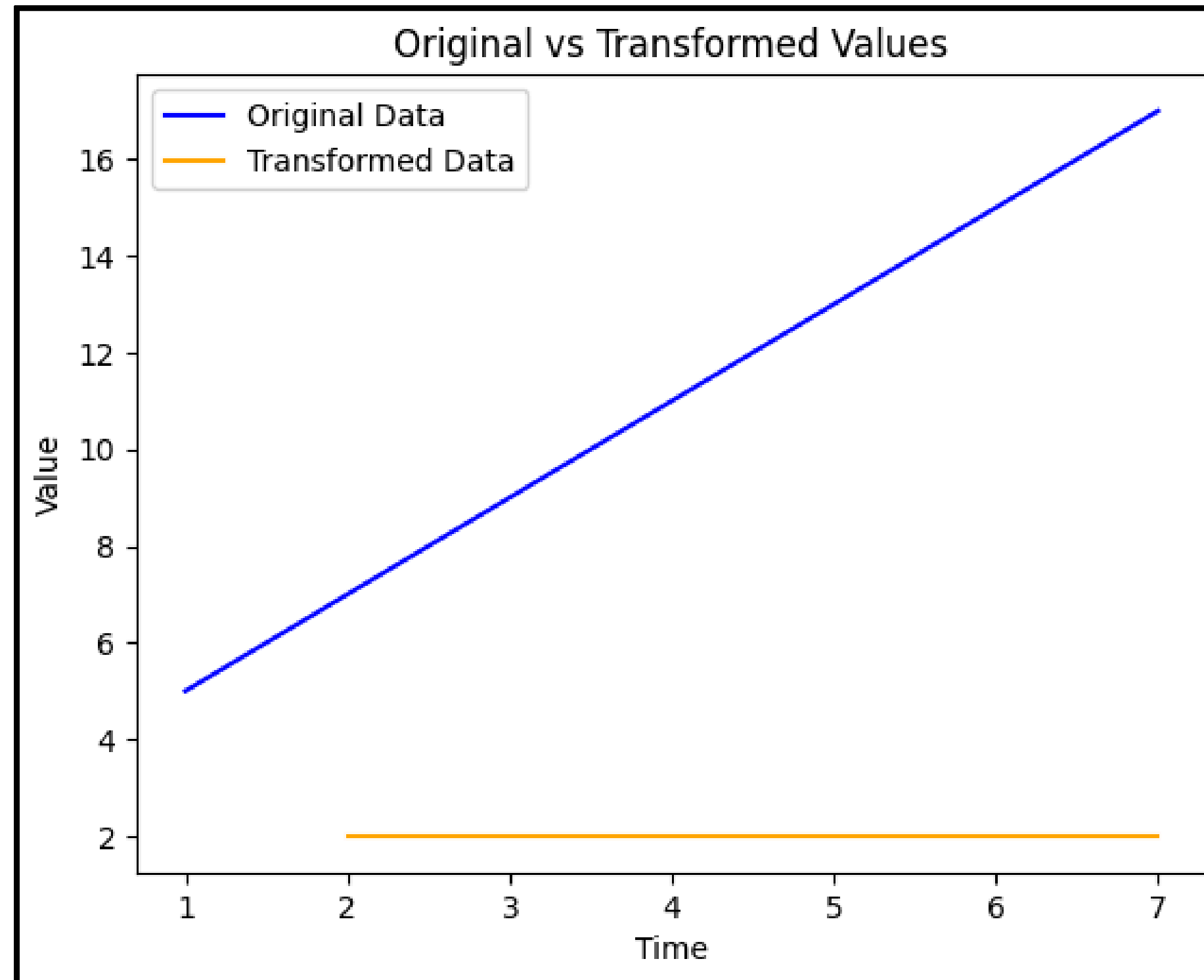
ΔX_t : **Change in Value**

This Produces a stable mean and variance.

Time (t)	Value	Lag 1	Transformed Values (Value - Lag 1)
1	5		
2	7	5	2
3	9	7	2
4	11	9	2
5	13	11	2
6	15	13	2
7	17	15	2

Difference Transformation

Plot of the previous tabular dataset



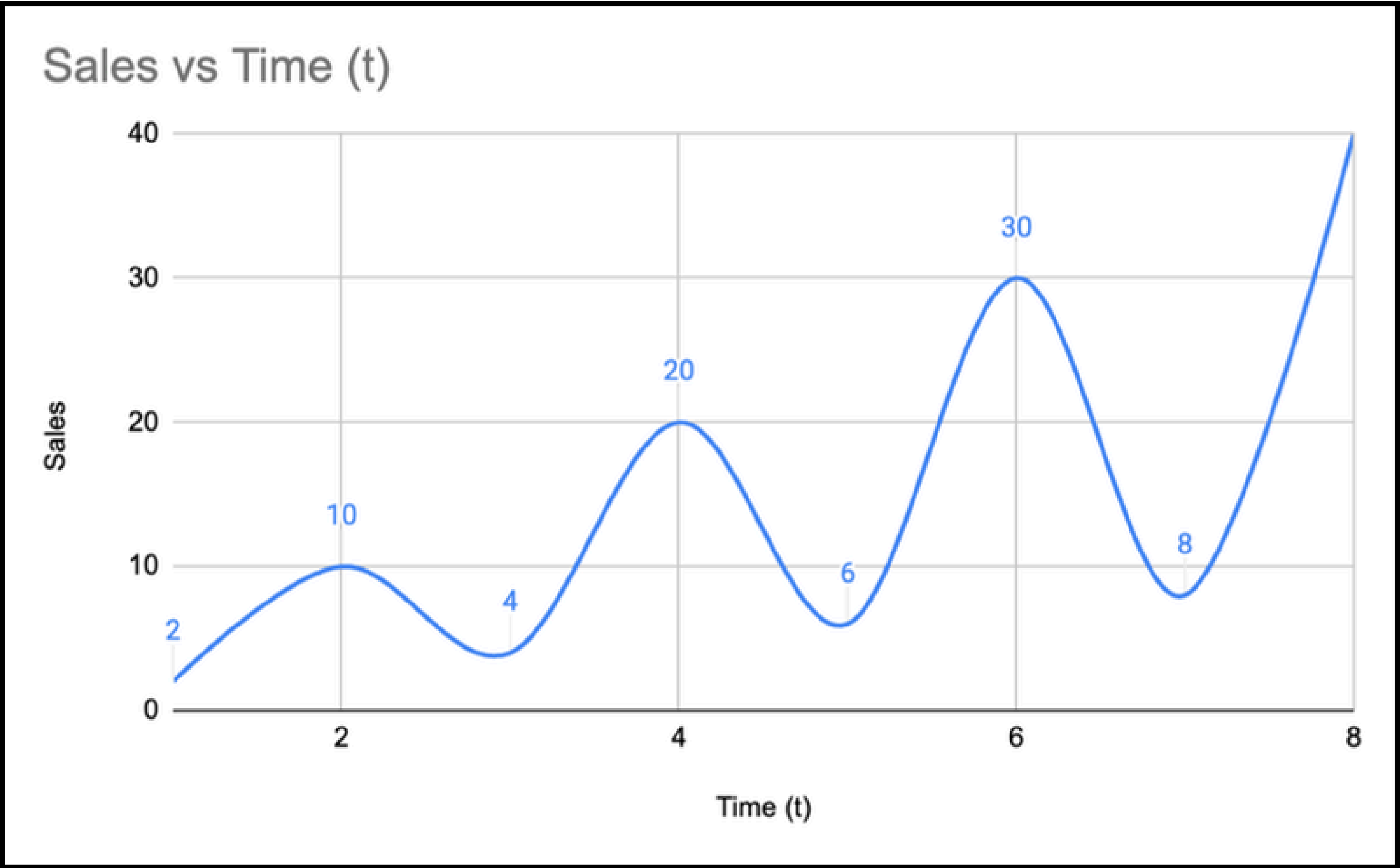
Can you analyze this data?

Time (t)	Sales
1	2
2	10
3	4
4	20
5	6
6	30
7	8
8	40

**Can difference transformation be
applied on it?**

Time (t)	Sales
1	2
2	10
3	4
4	20
5	6
6	30
7	8
8	40

Graph of Sales vs Time



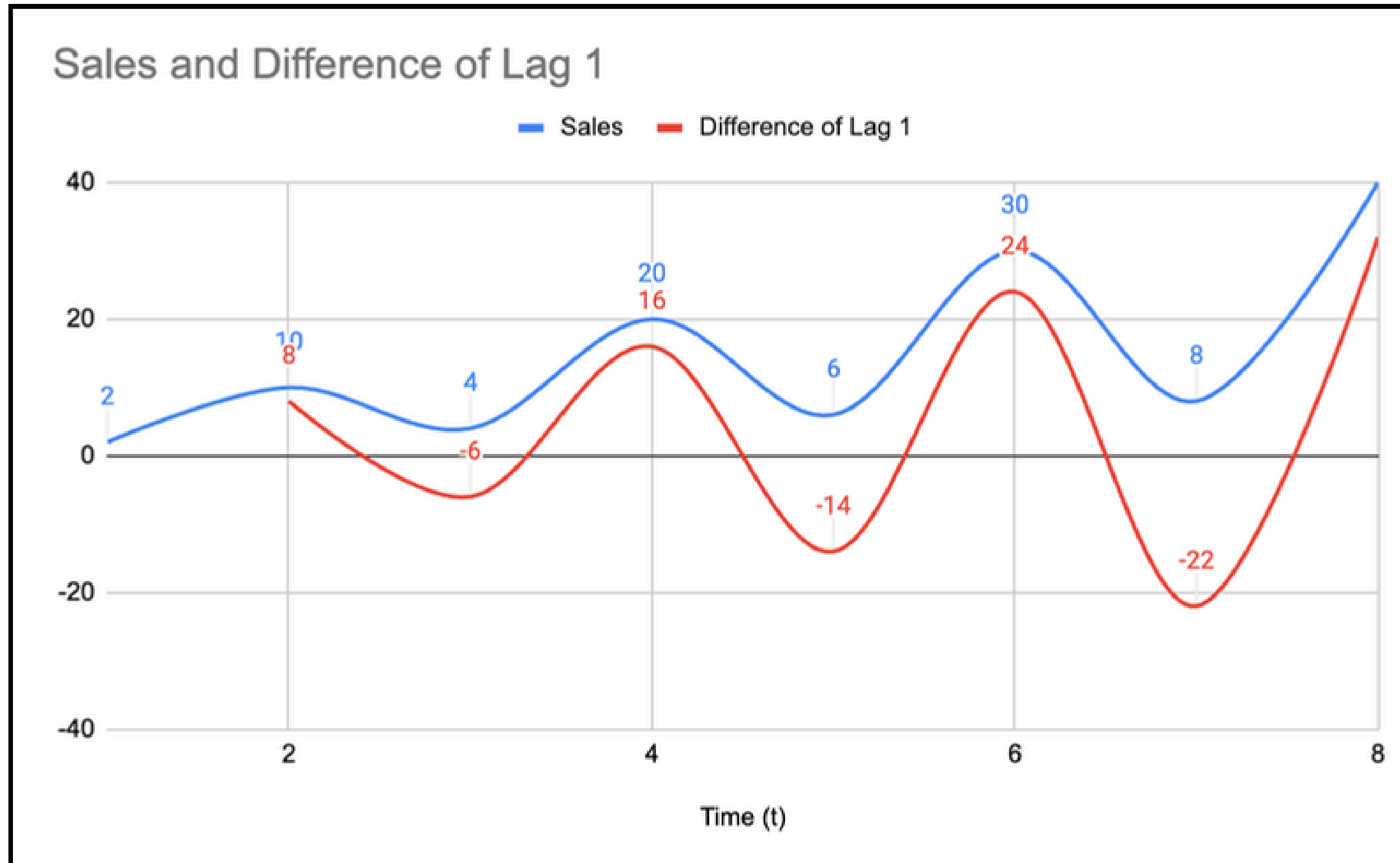
Time (t)	Sales
1	2
2	10
3	4
4	20
5	6
6	30
7	8
8	40

Non Stationary Data

Applying difference transformation on it.

Time (t)	Sales	Lag 1	Difference (Sales - Lag 1)
1	2		
2	10	2	8
3	4	10	-6
4	20	4	16
5	6	20	-14
6	30	6	24
7	8	30	-22
8	40	8	32

Applying difference transformation on it.



Applying difference transformation on it.

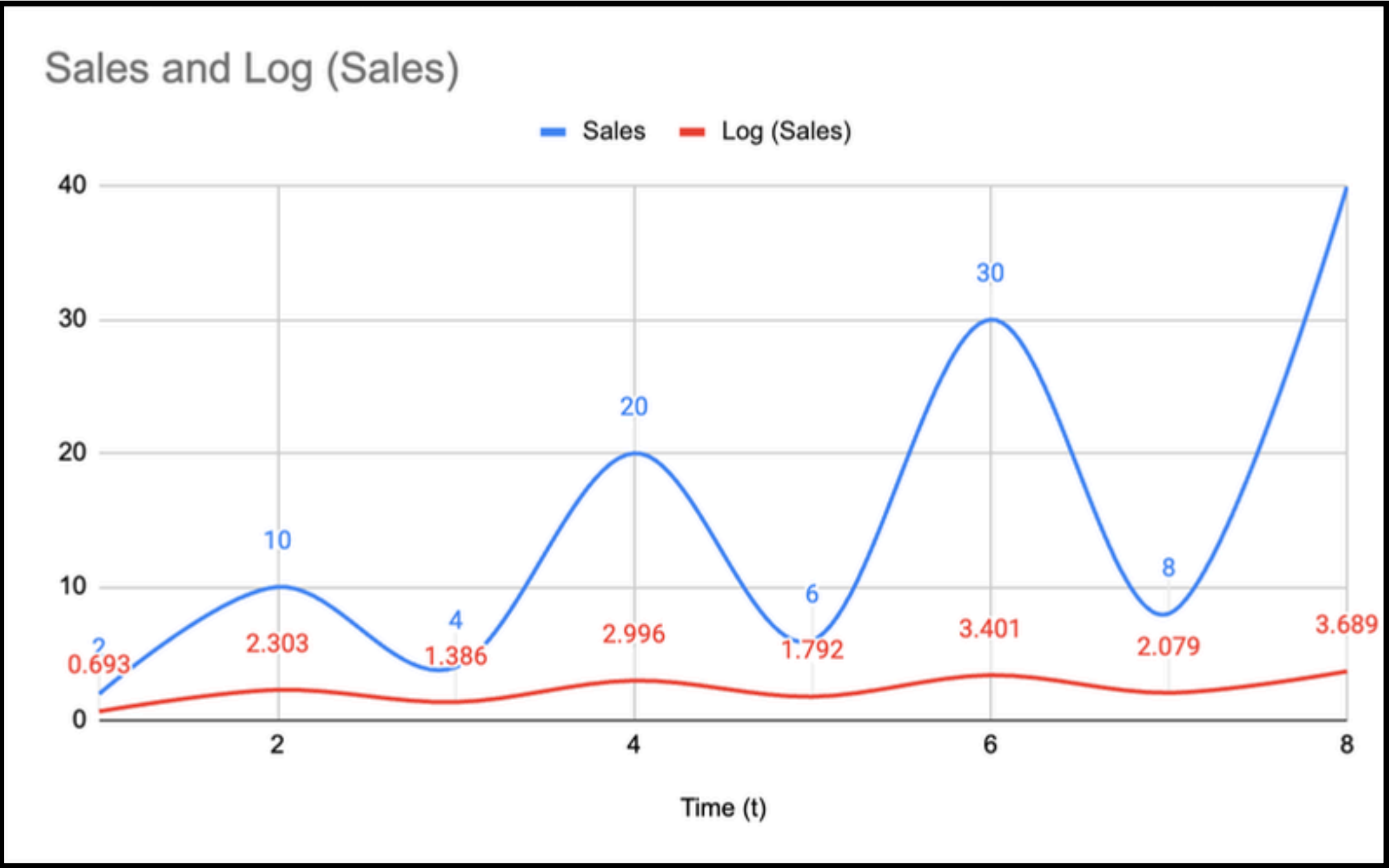
Why Differencing Fails

- Differences become larger as values grow
- Same pattern (jump up, fall down) but magnitude keeps increasing
- Variability is not stable
- Large values dominate the series

Key Insight: The problem is not only the change over time, but also that the change depends on how big the value already is

Time (t)	Sales	Lag 1	Difference (Sales - Lag 1)
1	2		
2	10	2	8
3	4	10	-6
4	20	4	16
5	6	20	-14
6	30	6	24
7	8	30	-22
8	40	8	32

Log Transformation



Time (t)	Sales	Transformed Values (Log (Sales))
1	2	0.693
2	10	2.303
3	4	1.386
4	20	2.996
5	6	1.792
6	30	3.401
7	8	2.079
8	40	3.689

Log Transformation

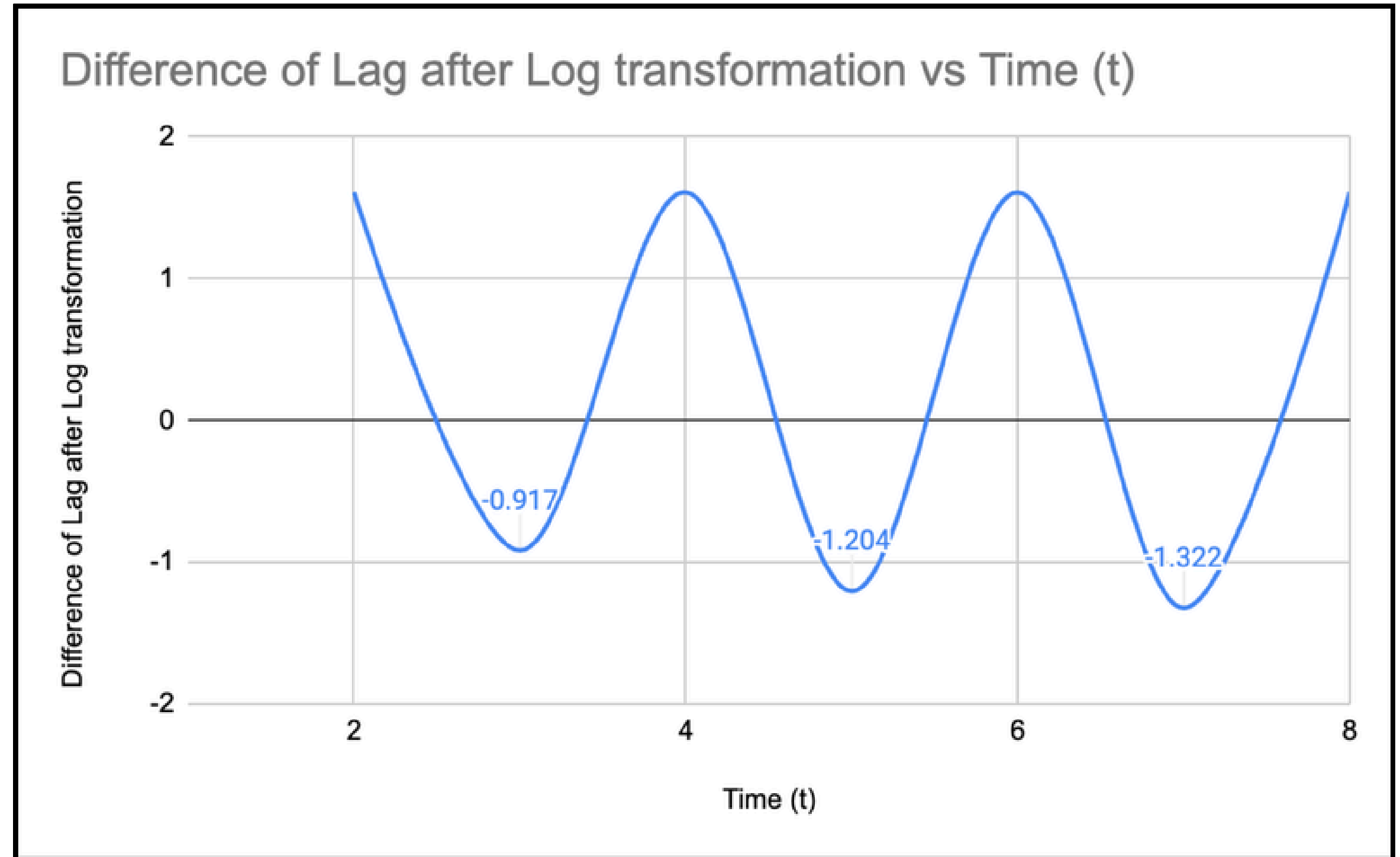
- Reduces **large fluctuations** by **compressing big** values.
- Stabilizes **variance**, especially when **ups and downs** grow over time.
- Makes **exponential growth** look more **linear** and easier to model.
- Helps **reveal underlying patterns** that were hidden by **big swings**.

$$Y_t = \log (X_t)$$

Time (t)	Sales	Transformed Values (Log(Sales))
1	2	0.693
2	10	2.303
3	4	1.386
4	20	2.996
5	6	1.792
6	30	3.401
7	8	2.079
8	40	3.689

Lag Difference after Log Transformation

Time (t)	Log(Sales)	Lag 1	Difference
1	0.693		
2	2.303	0.693	1.61
3	1.386	2.303	-0.917
4	2.996	1.386	1.61
5	1.792	2.996	-1.204
6	3.401	1.792	1.609
7	2.079	3.401	-1.322
8	3.689	2.079	1.61



Stationary Data

Comparison of all of the graphs in one image



Autocorrelation

- **What is Autocorrelation?**

- Autocorrelation is a mathematical concept to check the dependency of today's values with past values.
- In simple terms: to check whether the past influences the present

- **How does it help?**

- Detect patterns + seasonality (weekly cycles)
- Build better forecasts
- Avoid assuming the data is random

Mathematical formula for Autocorrelation

Autocorrelation at lag k :

$$\rho(k) = \text{Corr}(x_t, x_{t-k})$$

Expanded form:

$$\rho(k) = \frac{\sum_{t=k+1}^N (x_t - \mu)(x_{t-k} - \mu)}{\sum_{t=1}^N (x_t - \mu)^2}$$

Where:

- N = number of observations
- μ = mean of time series
- k = lag

Please fill the feedback form.

Thank You